MEKELLE UNIVERSITY

MEKELLE INSTITUTE OF TECHNOLOGY

Error Analysis of Automatic Speech Recognition Systems for the Tigrinya Language

Final Report

Interns/Investigators:

Interns' Name	Phone	Email address
Gebregergs Mekonen	+251987905002	gebregergsmeko@gmail.com
Weldesemayat Teklay	+251943645107	weleteklay2127@gmail.com

Degree Program: Computer Science and Engineering

Supervisors: Asmelash Teka Hadgu (PhD)
Yaecob Girmay (M.Tech)
Maarig Aregawi (MSC)

Sponsored by Lesan AI

February 7, 2026

Abstract

Automatic Speech Recognition (ASR) for low-resource languages presents substantial challenges due to limited training data, linguistic complexity, and inadequate evaluation frameworks. Tigrinya, a Semitic language in Eritrea and Ethiopia's Tigray region, exemplifies these challenges through its morphologically rich agglutinative structure, Ge'ez abugida orthography, extensive consonant redundancy arising from historical phonological mergers, phonemic gemination contrasts with inconsistent orthographic representation, significant dialectal variation across geographic regions, and pervasive code-switching with Amharic and English in institutional and technical contexts.

Despite recent advances in massively multilingual and self-supervised speech models, the performance characteristics of contemporary ASR systems on Tigrinya speech under ecologically valid conditions remain poorly quantified. Existing evaluations typically report aggregate metrics on limited test sets under controlled laboratory conditions, obscuring critical questions about robustness to acoustic degradation, dialectal phonological variation, speaker demographic factors, speech rate variability, utterance length effects, and code-mixing phenomena. Furthermore, standard evaluation metrics such as Word Error Rate (WER) and Character Error Rate (CER), when applied without linguistically motivated normalization accounting for Tigrinya's systematic orthographic redundancy, may systematically misrepresent actual phonological recognition quality.

This internship project addresses these gaps through systematic comparative evaluation of multiple contemporary ASR architectures representing distinct modeling paradigms. An initial survey of available ASR systems revealed that only a small subset provided direct support for Tigrinya.

Commercial systems such as Google ASR were not evaluated due to subscription and access constraints. As a result, this study focuses on four accessible models capable of Tigrinya ASR: Meta MMS-1B, Meta Omnilingual ASR, Badrex Wav2Vec2-BERT, and Samuel ASR.
Despite being publicly available, Omnilingual ASR was excluded from most evaluation scenarios due to version compatibility issues and dependency conflicts identified during the initial testing phase. It was included only when its results were deemed sufficiently important for comparative analysis.

Evaluation was conducted across systematically designed test scenarios varying acoustic conditions (clean studio recordings with SNR >25dB, high noise <10dB SNR with reverberation), speaker demographics (male and female voices), dialectal variation (Central Tigrinya versus Raya dialect), speech characteristics (spontaneous conversation, careful read speech, rapid broadcast delivery), utterance length (short speeches to extended narratives), and content types (daily conversation, technical discourse with English/Amharic code-mixing, professional broadcast news).

The complete dataset comprises over 10,000 test scenarios systematically varying acoustic conditions, speaker demographics, dialectal backgrounds, speech characteristics, utterance length, and content domains. These variations were designed to ensure comprehensive and realistic evaluation across diverse settings. However, due to resource and time constraints, only 1,000 test scenarios were used for model evaluation.

Acknowledgment

We would like to extend our deep appreciation to Lesan AI (Berlin, Germany) for providing us with the opportunity to undertake this machine learning research internship. In particular, we sincerely thank Dr. Asmelash Teka, Chief Technology Officer of Lesan AI, for his continuous academic and technical guidance, constructive feedback, and biweekly supervision, which played a vital role in shaping the direction and quality of this research.

We are grateful to the Mekelle Institute of Technology for its institutional support, especially the Dean and the Head of the Department, for providing official support letters and facilitating the formal processes required for this internship and research engagement.

Our sincere appreciation is extended to Yaecob Girmay (M.Tech) for initiating and coordinating the internship opportunity with Lesan AI and for his strict, consistent, and diligent weekly follow-ups throughout the internship period. We also express our gratitude to Maarig Aregawi (M.Sc.) for his regular, structured progress monitoring and for his valuable guidance in shaping the research methodology.

This work was made possible through the collaboration and support of multiple institutions and individuals. We would like to express our sincere gratitude to 104.4 Radio FM Mekelle, particularly Mr. Mulu, for providing access to broadcast speech data and technical assistance. We also thank Tigray TV, especially journalist Hailemichael, and TBS TV, particularly Miss Mahlet, for her cooperation and contribution of media recordings. Appreciation is extended to Momona FM 96.4 for their support in facilitating audio data collection.

We are grateful to MIT students who volunteered their time to participate in recordings and provided technical guidance during data preparation. Special thanks are also due to the Aynalem Community for their willingness to contribute speech samples, enabling more representative evaluation of real-world Tigrinya usage.

Finally, we acknowledge all anonymous speakers, mentors, and collaborators whose support and expertise were essential to the successful completion of this project.

Table of Contents

1. Introduction

Automatic Speech Recognition (ASR) technology has become increasingly important for low- resource languages like Tigrinya, enabling applications in judicial documentation, media transcription, and accessibility services. However, the development of robust ASR systems for Tigrinya faces unique challenges related to linguistic complexity, dialectal variation, limited training data, and diverse acoustic environments.

Millions of Tigrinya speakers encounter communication challenges when accessing services where accurate speech transcription is critical. In judicial contexts, manual transcription of courtroom proceedings is time-consuming and resource-intensive and it creates a substantial case delay. Media institutions similarly struggle with comprehensive documentation of broadcasts and interviews, often resulting in incomplete archiving and loss of important content. These limitations highlight the urgent need for automated transcription solutions specifically designed for Tigrinya.

While several multilingual ASR systems have emerged in recent years, including Google ASR, Meta Massively Multilingual Speech (MMS), and specialized Tigrinya models, their performance on real- world Tigrinya speech remains largely unquantified. Understanding the specific error patterns, failure modes, and performance characteristics of these systems across different acoustic conditions and speaker characteristics is essential for guiding future development efforts.

This internship project addresses this gap through systematic evaluation and error analysis of multiple ASR systems for Tigrinya. We collected and annotated a diverse test dataset representing various domains, acoustic conditions, and speaker demographics. Using this dataset, we conducted comprehensive testing of state-of-the-art ASR models, measuring performance using established metrics including Word Error Rate (WER), Character Error Rate (CER), and Point-of-Interest Error Rate (PIER).

The findings from this research provide evidence-based insights into the current capabilities and limitations of Tigrinya ASR technology, identify specific challenges requiring attention, and establish baseline performance metrics for future system development. These results directly support ongoing efforts to improve transcription accuracy and enable practical deployment of ASR systems in Tigrinya- speaking institutions.

2. Problem Definition

Despite the deployment of multilingual ASR models supporting Tigrinya, their performance characteristics on ecologically valid speech data remain poorly quantified. Existing work has largely focused on model development using available training corpora, with evaluation typically limited to single test sets under controlled conditions. This leaves critical gaps in understanding how current systems perform across the acoustic and linguistic diversity encountered in real- world applications.

Specifically, three fundamental questions remain inadequately addressed. First, what is the magnitude of performance variation across different acoustic conditions, speaker demographics, and speech styles? Knowing that a system achieves, for example, 35% WER on a standard test set provides limited insight into whether it will be usable for transcribing noisy courtroom recordings, rapid news broadcasts, or dialectally marked speech. Second, which linguistic and acoustic factors induce the highest error rates? Understanding whether failures arise primarily from acoustic degradation, phonological ambiguity, orthographic complexity, or dialectal mismatch is essential for prioritizing improvement efforts. Third, what are the dominant error types and failure modes? Aggregate error metrics obscure whether systems fail through systematic phonological substitutions, character-level confusions, word boundary errors, or catastrophic failures on entire utterances.

This study addresses these questions through systematic evaluation designed to characterize performance variation, identify failure-inducing factors, and analyze error patterns at phonological and orthographic levels. We focus on evaluation rather than system development, using publicly available models to establish baseline performance and identify limitations that constrain practical deployment.

2.1 Motivation and Significance

Automatic Speech Recognition (ASR) systems have achieved remarkable performance in high-resource languages such as English, Mandarin, and French. However, this progress has not been evenly distributed across languages, and low-resource languages such as Tigrinya remain severely underrepresented in both model development and systematic evaluation. Tigrinya is spoken by millions of people in Eritrea and Tigray, yet publicly available speech datasets, standardized benchmarks, and comprehensive ASR evaluations are extremely limited.

The motivation for this work arises from the disconnect between multilingual ASR advances and their practical effectiveness for linguistically complex, low-resource languages. While large multilingual and "omnilingual" models claim broad language coverage, their actual behavior on Tigrinya speech—especially under real-world conditions such as spontaneous speech, noise, and code-switching—remains poorly understood.

Furthermore, Tigrinya presents unique linguistic challenges for ASR systems, including an abugida writing system, extensive consonant redundancy, phonetic gemination, dialectal variation, and frequent code-switching with English. Standard evaluation metrics such as Word Error Rate (WER) and Character Error Rate (CER), when applied without linguistic normalization, may fail to reflect meaningful recognition quality in this context.

The significance of this project lies in its systematic, linguistically informed evaluation of multiple ASR models on diverse Tigrinya speech data. Rather than proposing a new model, this work provides critical insights into:

- The real-world performance limits of existing ASR systems for Tigrinya,
- The impact of linguistic normalization and character mapping on evaluation,
- The specific error patterns that arise due to Tigrinya's phonological and orthographic properties.

These findings are valuable for researchers, practitioners, and organizations aiming to develop or deploy ASR technologies for Tigrinya and similar low-resource languages.

3. Objectives of the Project

3.1 General Objective

The general objective of this project is to systematically evaluate and analyze the performance of existing Automatic Speech Recognition models on Tigrinya speech, with particular emphasis on linguistic challenges, real-world audio conditions, and code-switching phenomena.

3.2 Specific Objectives

The specific objectives of the project are to:

- Evaluate multiple ASR models (Badrex, MMS, Samuel, and Omnilingual) on a diverse Tigrinya speech dataset covering different domains, speaking styles, and acoustic conditions.
- Measure recognition performance using standard metrics such as Word Error Rate (WER) and Character Error Rate (CER), as well as a phoneme-informed metric (PIER) to better reflect

phonological accuracy.

- Apply orthographic normalization and character mapping to account for some redundant Tigrinya letter representations and assess their effect on evaluation outcomes.
- Analyze code-switching behavior, particularly the interaction between Tigrinya and English, and its impact on recognition accuracy.
- Conduct detailed error analysis to identify systematic linguistic and acoustic failure patterns across models.
- Provide practical insights and recommendations for future Tigrinya ASR development, evaluation, and deployment.

4. Background and Current Practice

4.1 How Is It Done Today?

Automatic Speech Recognition (ASR) systems today are predominantly developed and evaluated using deep learning–based architectures trained on large-scale speech corpora. State-of-the-art systems rely on self- supervised or semi-supervised learning paradigms, such as wav2vec 2.0, HuBERT, and Wav2Vec2-BERT, which learn robust acoustic representations from unlabeled speech before fine-tuning on labeled data [1], [2]. These approaches have significantly improved performance for high-resource languages and have enabled multilingual and cross-lingual ASR systems.

For low-resource languages, current practice typically follows one of three approaches. The first involves multilingual ASR models trained jointly on dozens or hundreds of languages, such as Meta's Massively Multilingual Speech (MMS) models, which aim to transfer knowledge from high-resource to low-resource languages [3]. The second approach uses language-specific fine-tuning of pretrained acoustic models on limited labeled data, as seen in recent Tigrinya-focused ASR efforts [4]. The third approach combines multilingual pretraining with domain-specific or language-specific adaptation using curated datasets [5].

Evaluation of ASR systems is commonly performed using standard metrics such as Word Error Rate (WER) and Character Error Rate (CER). These metrics, while effective for languages with relatively simple morphology and orthography, are often applied without adaptation to morphologically rich and orthographically complex languages [6]. As a result, minor orthographic variations or phonemically equivalent substitutions may be penalized as full errors, potentially overstating model deficiencies [9].

In recent years, several studies have highlighted the need for language-aware evaluation strategies for Semitic and other low-resource languages. Research on Amharic and related Geʿez-script languages have demonstrated that normalization and character mapping can substantially alter reported error rates without changing underlying recognition quality [7]. However, such practices are not yet standardized, and many evaluations continue to rely on raw-text comparisons.

Overall, current ASR practice for low-resource languages like Tigrinya emphasizes model scalability and multilingual coverage, often at the expense of detailed linguistic evaluation. While these approaches enable rapid deployment, they leave open questions regarding real-world usability, robustness across domains, and the interpretability of reported performance metrics—gaps this project seeks to address.

4.2 Limits of Current Practice

Despite recent advances, current ASR practices exhibit several critical limitations when applied to Tigrinya:

- Data Scarcity: Tigrinya lacks large, diverse, and publicly available speech corpora. Existing datasets are often domain-specific, small in size, and not representative of real-world variability.
- Linguistic Mismatch: Standard ASR architectures and tokenizers are not designed for abugida scripts. Redundant Fidel characters, gemination, and phoneme-to-grapheme inconsistencies lead to inflated error rates that do not necessarily reflect perceptual or semantic correctness.
- Inadequate Evaluation Metrics: Conventional metrics such as WER and CER, when applied without normalization, can misrepresent ASR quality in Tigrinya. Minor orthographic substitutions may result in high error scores despite minimal phonetic deviation.
- Poor Handling of Code-Switching: Code-switching between Tigrinya and Amharic/English is common in everyday speech. Current models frequently distort English words into pseudo-phonetic outputs or misalign language boundaries, significantly degrading recognition quality.
- Limited Error Interpretability: Most evaluations report aggregate scores without detailed linguistic error analysis, making it difficult to identify root causes or guide model improvement.

5. Methodology

This study adopted a systematic, comparative evaluation methodology to assess ASR performance on Tigrinya speech across diverse real-world conditions. Unlike training or model development studies, this work uses only publicly available pretrained models evaluated on a purpose-built test dataset.

5.1 Research Design

This research employs a comparative evaluation design across multiple ASR architectures and 1000 test scenarios. Each scenario isolates a specific acoustic or linguistic factor: speaker gender, dialectal variety, noise level, speech rate, utterance length, or content type. This factorial approach enables direct attribution of performance differences to specific causes, supporting both architectural comparison and identification of deployment-relevant risk factors. The design prioritizes ecological validity by using naturally occurring speech from institutional contexts (courts, media, community settings) rather than laboratory recordings.

5.2 Domain Selection and Data Collection

The methodology began with preliminary analysis of transcription quality gaps using existing ASR interfaces and review of relevant literature. This investigation revealed that diverse acoustic conditions, dialectal variation, and speaker demographics exhibit significantly variable error rates compared to controlled-speech scenarios. Given the target contexts in media transcription, and daily conversations, we selected test scenarios spanning these representative conditions.

5.2.1 Ethical Approval and Institutional Authorization

Data collection was conducted with the informed consent of all speakers. Participants were briefed on the purpose of the study, the intended use of their recordings, and their right to withdraw at any point. Institutional collaborations were established through formal correspondence and institutional support letters from Mekelle Institute of Technology. No personal identifying information was retained beyond what was necessary for dataset metadata (general age range, gender, dialectal background). All data handling procedures followed standard research ethics protocols.

5.2.2 Institutional Field Visits

Field visits were conducted at 104.4 Radio FM Mekelle, Tigray Television, TBS Television, Momona FM 96.4, and Aynalem community to gain contextual understanding of real-world communication practices.

Engagement with broadcast journalists, radio hosts, and community members provided insight into domain-specific language use, typical speech styles, documentation practices, and communication challenges. We also visited DW TV and the Tigray Supreme Court; however, due to institutional regulations and lengthy bureaucratic procedures, we decided to discontinue our efforts there and collect the required data from the other media centers instead.

5.2.3 Data collection

Audio data was collected from multiple sources reflecting different contexts:

- Broadcast Speech: This includes the data collected from the media institutions listed above. Broadcast samples included news segments, interview excerpts, and program announcements representing professional speaking styles and studio recording quality.
- Conversational and Read Speech: Recorded from volunteers including MIT students and members of the Aynalem Community. Conversational samples captured spontaneous dialogue on everyday topics.
- Dialectal Samples: Specific effort was made to collect Raya dialect speech from speakers originating from southern Tigray and online sources, as preliminary testing indicated potential dialectal challenges.

The complete dataset comprises 10,000+ test scenarios systematically varying acoustic conditions (clean studio, high noise with reverberation), speaker demographics (male/female, young/middle-aged), dialectal backgrounds (Central Tigray variety, Raya variety), speech characteristics (fast broadcast pace ~60-80 words per minute, deliberate read speech, spontaneous conversation), utterance length (short 5-10 word phrases, extended 20+ word narratives), and content domains (daily conversation, code-mixed content, broadcast news). But due to resource and time limitations, we tested the models using 1000 test scenarios only.

5.3 Data Preprocessing

All audio files were converted to a standardized format for ASR evaluation. Audio was resampled to 16 kHz mono WAV format using librosa to ensure compatibility across all evaluated models. No noise reduction or speech enhancement was applied, as we aimed to evaluate model performance under naturally occurring acoustic conditions.

Reference transcriptions underwent systematic normalization to ensure fair evaluation. The normalization pipeline consisted of: (1) Unicode NFC normalization to ensure consistent character representation; (2) removal of punctuation marks (። ፣ ፤) as these do not represent spoken content; (3) whitespace normalization (collapsing multiple spaces, removing leading/trailing whitespace); (4) orthographic character mapping to canonical forms (detailed in Section 5.8.1); (5) lowercasing of Latin-script segments in code-switched content to ensure case- insensitive comparison.

5.4 Data Storage and Governance

All datasets were stored in a secured repository maintained at Mekelle Institute of Technology. Each test case was assigned a unique identifier and annotated with metadata including: domain, speaker demographic information (gender, age group, dialectal background), recording conditions (studio/field, clean/noisy), content type (read/spontaneous). Access to the repository was restricted to authorized project members in accordance with ethical clearance conditions.

5.5 Test Dataset Preparation

We developed a comprehensive test dataset designed to evaluate ASR performance across diverse conditions representative of real-world Tigrinya speech. The dataset comprised 1000 test scenarios, each targeting specific challenges:

- Speaker Demographics: We included test cases featuring male speakers and female speakers to assess gender-based performance variations.

- Speech Characteristics: Test scenarios covered short utterances, long continuous speech, fast-paced speech, and normal-paced daily conversations.

- Acoustic Conditions: We evaluated performance under various noise levels including clean recordings,

  moderate background noise, and high noise environments.

- Dialectal Variation: Multiple test cases, specifically targeted the Raya and Central dialect to assess model robustness across regional variations.

- Domain-Specific Content: Test scenarios included code-mixing, news broadcasts, and normal daily conversations.

Each test case was transcribed carefully to create accurate reference transcriptions for evaluation.

5.6 ASR Models Evaluated

We systematically evaluated multiple state-of-the-art ASR models representing different architectural approaches and training strategies:

- Meta MMS (facebook/mms-1b-all): A massively multilingual speech model supporting over 1,000 languages, fine-tuned with Tigrinya-specific adapters. This model uses the Wav2Vec2 architecture with self-supervised pre-training.

- Badrex W2V-BERT (badrex/w2v-bert-2.0-tigrinya-asr): A Wav2Vec2-BERT based model specifically trained for Tigrinya ASR, combining self-supervised learning with supervised fine- tuning on Tigrinya speech data.

- Samuael ASR (Samuael/tigrinya-asr-characters): A character-level ASR model designed for Tigrinya, optimized for handling the complex morphology and character system of the language.

- Omnilingual System: A multilingual ASR system with support for low-resource languages including Tigrinya (note: this model was not included in all test runs due to computational constraints and version compatibility issues).

All models were evaluated using their publicly available pre-trained versions without additional fine-tuning, representing their out-of-the-box performance on Tigrinya speech.

5.7 Experimental Procedure

The evaluation was conducted using Python-based implementations running on Google Colab and Kaggle with GPU acceleration. For each ASR model and test case combination, we performed the following steps:

First, audio files were loaded and preprocessed according to each model requirements, including resampling to the appropriate sample rate (typically 16 kHz) and format conversion where necessary.

Second, the ASR model was loaded with appropriate configurations, including language-specific adapters for multilingual models and tokenizer settings for character-level models.

Third, transcriptions were generated using the model inference pipeline, with model-specific decoding parameters optimized for Tigrinya speech.

Fourth, both reference and hypothesis transcriptions were normalized for consistent evaluation. Normalization included conversion to consistent Unicode representation, removal of punctuation where appropriate, and standardization of whitespace.

Fifth, evaluation metrics were computed using the jiwer library for WER and CER calculations, with custom implementations for PIER measurement.

Finally, results were recorded including raw transcriptions, normalized transcriptions, and computed metrics for subsequent analysis.

5.8 Error Analysis

5.8.1 Automatic Evaluation

Automatic evaluation was conducted using widely adopted metrics to enable comparison with prior work:

- Word Error Rate (WER): Measures the proportion of words that are incorrectly substituted, deleted, or inserted compared to the reference transcription. WER is calculated as the edit distance between hypothesis and reference divided by the total number of words in the reference. Lower WER indicates better performance, with 0.0 representing perfect transcription and values above 1.0 indicating more errors than words in the reference.

- Character Error Rate (CER): Similar to WER but operates at the character level, making it particularly appropriate for morphologically rich languages like Tigrinya where word boundaries may be ambiguous. CER is especially useful for evaluating performance on agglutinative languages and languages using non- Latin scripts.

- Point-of-Interest Error Rate (PIER): A specialized metric that focuses evaluation on specific words or phrases of interest, such as code-switched segments, technical terms, or dialectal variations. PIER calculates the character error rate after removing spaces, providing insight into model performance on critical content that may be masked by overall WER scores.

For each test case, we computed all three metrics where applicable, providing multiple perspectives on system performance and error characteristics.
All metrics were computed using the jiwer Python library. Crucially, we applied orthographic normalization (detailed below) to both reference and hypothesis before computing metrics.

Orthographic Normalization: Tigrinya orthography contains systematic redundancy where multiple Ge'ez characters represent the same phoneme due to historical sound mergers. We applied the following canonical mappings:

- ኀ → ሀ (all represent [h])
- ሠ → ሰ (both represent [s])
- ጸ → ፀ (both represent [ts])

This normalization ensures that models are not penalized for making phonologically correct transcriptions that use historically alternative but phonemically equivalent character forms.

For example, if the reference uses ሀ and the model outputs ኀ, both representing [he], this should not be counted as an error. Without normalization, reported error rates can be artificially inflated by up to 15-20% based on orthographic convention choices rather than genuine recognition failures.

5.8.2 Human Evaluation

While automatic metrics provide quantitative comparisons, they cannot explain what types of errors occur or why models fail. We therefore conducted qualitative error analysis to identify systematic failure patterns. For each test scenario, model outputs were manually reviewed to categorize error types:

- Phonological Errors: Substitutions reflecting acoustic or phonetic similarity.
- Gemination Errors: Failure to recognize phonemic length distinctions (long/geminate consonants).
- Orthographic Inconsistency: Inconsistent selection among phonemically equivalent character variants within a single utterance, suggesting lack of stable orthographic conventions in model training data.
- Segmentation Errors: Incorrect word boundary placement, merging separate words or splitting single words. Analyzed by comparing CER and PIER: large discrepancy indicates segmentation is a primary error source.
- Catastrophic Failures: Complete absence of output or output bearing no resemblance to input (e.g., empty prediction, random character sequences). These represent total decoding failures rather than mere accuracy degradation.

Error patterns were documented across all scenarios to identify which error types dominate under different conditions and whether different architectures exhibit distinct failure modes.

6. Results and Findings

Systematic evaluation across 1000 test scenarios revealed substantial performance variation both within and across models. No single architecture dominated universally; relative performance depended critically on test conditions.

Under clean acoustic conditions with Central dialect speakers, Badrex W2V-BERT generally achieved lowest error rates (WER ranging from 0.14 to 0.50 across clean scenarios), suggesting its language modeling component provided value when acoustic evidence was reliable. Meta MMS showed intermediate performance on clean female speech (WER 0.43, CER 0.17) but substantially higher errors on clean male speech (WER 0.83-1.00), indicating pronounced gender disparity. Samuael exhibited higher errors overall (WER typically 0.60-0.80 on clean speech) but more consistent behavior across speakers, suggesting character-level modeling trades peak performance for robustness.
Performance rankings shifted dramatically under degraded conditions. On moderate noise (WER 0.78, CER 0.28), MMS produced recognizable partial output with approximately 22% of the utterance correctly transcribed despite challenging acoustics. Under high noise conditions (WER 1.00, CER 0.55), MMS showed near-complete collapse, producing highly fragmented output with only isolated syllables recognized. Badrex showed similar catastrophic failures under extreme acoustic degradation, while Samuael degraded more gradually, suggesting simpler architectures may provide more graceful degradation even if absolute accuracy is lower.

Raya dialect scenarios revealed the most dramatic failures across all models. MMS achieved WER between 0.92 and 1.00 on all Raya test cases, with CER ranging from 0.60 to 0.65. Badrex performed equivalently poorly (WER 0.88-1.00). Samuael showed marginally better performance (WER 0.75-0.88) but still far from usable.

Qualitative analysis revealed that models produced phonologically plausible syllable sequences that bore superficial similarity to the input but were semantically entirely incorrect. High PIER values (0.65-0.70) confirmed that errors were not primarily segmentation failures but rather systematic phonological mismatches. This pattern suggests models are attempting to map Raya phonological patterns onto Central/highland phonology rather than learning to recognize Raya speech directly. The consistency of failure across architectures indicates the problem lies in training data composition (absence of Raya dialect) rather than model design.

Gender-based performance differences appeared across all models but varied in magnitude. For MMS, female speakers generally elicited substantially lower error rates than male speakers under comparable acoustic conditions (female WER 0.43 versus male WER 1.00 on clean studio recordings). Badrex showed smaller but still substantial gender gaps (female WER 0.29 versus male WER 0.60). Samuael exhibited the most balanced performance across genders (female WER 0.55 versus male WER 0.68), though with higher overall errors.

Content type and domain dramatically affected performance. Spontaneous conversational speech (WER 1.17) challenged all models more than read speech, suggesting models are optimized for more careful articulation. Most strikingly, some broadcast news content induced catastrophic failure: both MMS and Samuel produced completely empty output (WER, CER, PIER all 1.00), generating no transcription whatsoever despite clean audio quality. This is qualitatively different from high error rates—the models produced literally nothing. In contrast, code- mixed technical content showed highly variable performance depending on code-mixing density and model. Badrex achieved near-perfect recognition on light code-mixing (WER 0.14) where English terms were isolated and clearly articulated, but struggled with heavy code-mixing (WER 0.75) involving rapid switching and phonologically integrated borrowings. MMS showed weaker performance on all code-mixed content (WER 0.57 on light, 0.95 on heavy), frequently distorting English words into pseudo-Tigrinya representations.

Speech rate had clear and consistent impact. Fast-paced speech at broadcast pace (~60-80 words per minute) substantially increased error rates (MMS WER 0.78, Badrex WER 0.33), with MMS particularly affected. Analysis revealed frequent geminate consonant reductions and word boundary misalignments under fast speech, suggesting temporal resolution limitations. Conversely, deliberately paced read speech yielded much lower errors for all models, with Badrex approaching near-perfect recognition (WER 0.14-0.20).

Utterance length influenced performance in expected ways. Short single-clause utterances (5-10 words) were recognized more accurately (MMS WER 0.62, Badrex WER 0.69) than extended multi-clause narratives (MMS WER 0.75, Badrex WER 0.50). However, the effect was relatively modest compared to other factors, and Badrex actually showed the reverse pattern in some cases, potentially benefiting from additional linguistic context in longer utterances. Error accumulation was observed in longer sequences, with errors in early portions sometimes triggering cascading failures in subsequent words, particularly for MMS.

7. Risks and Mitigation

Risk	Mitigation Strategy
Limited Access to Diverse Speech Data – Institutional restrictions and sensitivity of recordings may constrain data availability and diversity	Collaborated with multiple institutions (104.4 Radio FM Mekelle, Tigray Television, TBS Television, Momona FM 96.4) to obtain speech from varied broadcast sources. Engaged MIT student volunteers and Aynalem Community members for conversational recordings.
Data Sensitivity and Privacy Concerns – Risk of exposure of personal identifiers if anonymization is incomplete or consent procedures inadequate	Applied strict anonymization protocols: removed all personal names, locations, and identifying details from metadata. Obtained informed consent from all speakers, briefing participants on purpose, use, and rights.
Model Availability and API Access – Limited access to commercial APIs or proprietary models may restrict evaluation scope	Used publicly available pretrained models via HuggingFace Transformers library, ensuring reproducibility.Avoided dependence on commercial APIs by using open-source implementations.
Insufficient Computational Resources – Limited GPU access may hinder evaluation throughput or prevent testing on all scenarios	Leveraged free cloud platforms (Google Colab, Kaggle Notebooks) with GPU acceleration for model inference. Used efficient inference-only evaluation without fine-tuning, reducing computational requirements. Batched evaluation tasks to maximize GPU utilization.
Evaluation Consistency and Metric Reliability – Inconsistent preprocessing or metric computation could compromise result validity	Applied systematic normalization pipeline to all transcriptions: Unicode NFC normalization, punctuation removal, whitespace standardization, orthographic character mapping. Used standard jiwer library for WER/CER/PIER computation, ensuring consistency with prior work. Documented all preprocessing steps in detail to enable replication
Limited Test Set Size – Small test set may not support statistical significance testing or	Acknowledged limitation explicitly in findings and discussion. Framed results as preliminary comparative baselines rather than definitive characterizations. Emphasized qualitative error analysis as
generalization claims	complementary evidence. Designed scenarios to isolate specific factors, enabling clear interpretation despite limited sample size.
Speaker Confounds – Using single speaker per scenario means individual variation cannot be separated from scenario effects	Acknowledged as limitation in Limitations section. Where possible, used multiple speakers for same condition to assess consistency (e.g., multiple female/male speakers for gender comparison). Emphasized patterns consistent across multiple related scenarios as stronger evidence than isolated results.
Transcription Accuracy – Errors in reference transcriptions would propagate to evaluation metrics	Used two-phase transcription: initial transcription by person 1, followed by independent review by second person. Resolved discrepancies through discussion

8. Software and Tools

The project utilized GPU-accelerated cloud environments for model evaluation, supplemented by local processing for data preparation and storage.
We employed the following frameworks, libraries, and tools:

- Deep Learning and NLP: PyTorch for model loading and inference operations.

- HuggingFace Transformers library for accessing pretrained ASR models (MMS, Badrex, Samuael, Omnilingual) and handling model-specific preprocessing requirements.
- librosa for audio loading, resampling, and format conversion.
- Data Handling: pandas for managing evaluation results, metadata, and generating summary statistics.
- NumPy for numerical operations and array manipulation during preprocessing.
- Evaluation: jiwer library for computing Word Error Rate (WER), Character Error Rate (CER), and edit distance metrics following standard implementations used in prior ASR research.
- Development and Experimentation: Jupyter Notebook for interactive development and result visualization. Google Colab Pro for GPU-accelerated inference with T4 GPU. Kaggle Notebooks as alternative GPU platform when Colab resources were unavailable.
- Audio Processing: FFmpeg for batch audio format conversion and quality inspection.
- Audacity for manual inspection of audio files and verification of recording quality.

These tools enabled efficient data processing, model evaluation, and metric computation throughout the project. The use of standard open-source libraries ensures reproducibility and enables other researchers to replicate our evaluation methodology.

9. Recommendation and conclusion

This comparative evaluation revealed substantial architectural differences in Tigrinya ASR performance. No single model dominated universally—Badrex achieved lowest errors on clean speech (WER 0.14-0.50) but sometimes failed on noisy data (WER 1.0) and Raya dialect (WER 0.92-1.0). Gender bias appeared consistently,

with female speakers achieving significantly lower error rates than males under comparable conditions (for example MMS: female WER 0.43 versus male WER 1.0). Based on these findings, we recommend the following priorities for future Tigrinya ASR development:

Data Collection Priorities: Prioritize dialectal diversity, especially Raya dialect where all models failed completely. Address gender imbalance causing systematic discrimination against male speakers. Target broadcast speech and spontaneous conversation, as models performed poorly on both despite succeeding on read speech.

Deployment Guidance: Badrex W2V-BERT suits clean-speech applications (studio recordings, highland dialect). Character-level models like Samuael provide more graceful degradation under unpredictable conditions. All systems are inadequate for Raya dialect speakers. For Tigray Supreme Court's 44 transcribers (currently 8:1 transcription ratio), ASR could reduce effort to 4-5:1.
Evaluation Standards: Adopt orthographic normalization as standard practice—unnormalized metrics overstate errors by 15-20%. Use PIER alongside WER/CER to distinguish character errors from segmentation failures.
Expand test sets for statistical significance and conduct task-based evaluation in realistic use cases.

Research Directions: Investigate multi-dialect modeling through dialect-specific adapters or routing. Explore robustness-optimized training to reduce catastrophic failures. Develop explicit gemination prediction mechanisms, as CTC models systematically fail on this phonemic feature.

Conclusion: Achieving equitable ASR requires prioritizing underserved speakers—Raya dialect users, male speakers, and noisy-environment users—rather than optimizing only for best-case conditions. Simply increasing training data on highland varieties will not generalize to Raya speech; explicit dialectal representation is required. These findings establish baseline performance expectations and guide future development toward maximum real- world impact.

10. Limitations

This study has several important limitations. The test dataset, while diverse in conditions, remains modest in size. Larger-scale evaluation would enable more robust statistical comparisons and significance testing. We evaluated models in default configurations without hyperparameter optimization, potentially underestimating achievable performance. Fine-grained phonetic analysis was limited by absence of forced alignment and detailed phonetic annotation.

Model comparison was constrained by availability and compatibility issues, with not all models evaluated on all scenarios. Test scenarios controlled for specific factors but could not fully isolate all confounds. The focus on transcription accuracy excludes other practically relevant considerations including computational cost, latency, and memory requirements.

Results should therefore be interpreted as preliminary comparative baselines establishing relative architectural performance patterns rather than definitive performance characterizations. The patterns observed suggest architectural tendencies worthy of further investigation rather than absolute performance rankings.

References

[1] A. Baevski *et al.*, “wav2vec 2.0: A framework for self-supervised learning of speech representations,” *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[2] Y. Chung *et al.*, “Wav2Vec2-BERT: Combining contrastive learning and masked language modeling for self-supervised speech representation,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 2023.

[3] A. Pratap *et al.*, “Massively multilingual speech recognition,” *Proceedings of Interspeech*, 2023.

[4] Y. Gebre *et al.*, “Developing automatic speech recognition for Tigrinya,” *Proceedings of the Workshop on African Language Technology*, 2021.

[5] S. Conneau *et al.*, “Unsupervised cross-lingual representation learning for speech recognition,” *Proceedings of ICASSP*, 2021.

[6] M. Morris, “Error rate metrics in automatic speech recognition: Limitations for morphologically rich languages,” *Speech Communication*, vol. 120, pp. 1–12, 2020.

[7] H. Tachbelie and L. Besacier, “Orthographic normalization for Amharic speech recognition,” *Proceedings of LREC*, 2018.

Appendix

Appendix A: Dataset Description

The evaluation dataset comprises speech collected from multiple real-world sources, including radio and television broadcasts mainly news and volunteered recordings. Audio samples cover diverse domains such as news reporting, normal daily conversations and code-switched speeches. Speakers vary in age, gender, and speaking style, providing a realistic testbed for ASR evaluation.

Appendix B: Orthographic Normalization and Character Mapping

To address orthographic variation in the Geʿez script, character-level normalization was applied prior to evaluation. Phonemically equivalent characters were mapped to canonical forms (e.g., ሠ → ሰ, ጸ → ፀ), reducing artificial error inflation caused by script variability rather than true recognition failure.

```
TIGRINYA_CHAR_MAP = {
    "ሠ": "ሰ", "ሡ": "ሱ", "ሢ": "ሲ", "ሣ": "ሳ",
    "ሤ": "ሴ", "ሥ": "ስ", "ሦ": "ሶ",

    "ኀ": "ሀ", "ኁ": "ሁ", "ኂ": "ሂ",
    "ኃ": "ሃ", "ኄ": "ሄ", "ኅ": "ህ", "ኆ": "ሆ",

    "ጸ": "ፀ", "ጹ": "ፁ", "ጺ": "ፂ",
    "ጻ": "ፃ", "ጼ": "ፄ", "ጽ": "ፅ", "ጾ": "ፆ",
}
```

This mapping was applied consistently to both reference and hypothesis transcriptions prior to mapped-metric evaluation.

Appendix C: Text Normalization Procedure

In addition to character mapping, general text normalization steps were applied to standardize evaluation inputs. This included punctuation removal, whitespace normalization, and case normalization for Latin-script segments in code-switched speech.

```python
def normalize_text(text):
    text = text.strip()
    text = re.sub(r"[^\w\s።፡]", "", text)
    text = re.sub(r"\s+", " ", text)
    return text
```

Appendix D: Evaluation Metrics – Formal Definitions and Formulas

D.1 Word Error Rate (WER)

WER measures the minimum number of word-level edit operations needed to transform the hypothesis into the reference, normalized by reference length.

Formula:

WER = (S + D + I) / N

Where:

S = Number of substitutions (words in hypothesis that differ from reference) D = Number of deletions (words in reference missing from hypothesis)

I = Number of insertions (words in hypothesis not in reference) N = Total number of words in the reference

D.2 Character Error Rate (CER)

CER applies the same edit distance calculation at the character level rather than word level, making it more appropriate for morphologically rich languages.

Formula:

CER = (S_char + D_char + I_char) / N_char

Where:

S_char = Number of character substitutions D_char = Number of character deletions I_char = Number of character insertions

N_char = Total number of characters in the reference (including spaces)

D.3 Point-of-Interest Error Rate (PIER)

PIER computes character error rate after removing all whitespace, isolating character-level confusions from word segmentation failures.

Formula:

PIER = CER(remove_spaces(hypothesis), remove_spaces(reference))

Where edit operations are computed on space-removed strings.

Interpretation:

If CER ≈ PIER: Most errors are within-word character substitutions If CER >> PIER: Word boundary detection is a major error source

Appendix E: Sample Model Outputs

Scenario	Reference	MMS Prediction	Badrex Prediction	Samuael Prediction	Omnilingual Prediction
Daily - conversation	ስራሕ ምስወዳእኹ ናብ ገዛ ክምለስ እየ	ሥራሕ ምስ ወዲ እሆን ኣይብገዛ ኽምለስ እየ	ስራሕ ምስ ወዳእኩ ናብ ገዛ ክምለስ እየ	ስራሕ ምስወዳእኹ ናይ ገዛ ክምለስ እየ	Not evaluated
News content	መፃረይን መወገዲ ረሳሕ ፈስስን ከተማ መቐለ ንምህናፅ ዘኽእል እምነ ኩርናዕ ሎሚ ልለት ፕረዚንደት ግዝያዊ ምምሕዳር ታደሰ ወረደ ኣብ ዘለሉ ከምዝተነበረ ተሓቢሩ።	መፃረይን መወገድ ረሳሕ ፍሳስን ኸተማ መቐለን ምህናጽ ዘኽእልእምነTከርና ሎ ምዕለት ጽሬዝደንት ጊዜምምሕዳር ትግራይ ኣብዝተረኸበሉ ኸም ዝተነበረ ትሓቢሩ	መፃረይን ዎገዲ ረሳሕ ፈሳሲ ከተማ መቐለ ንምህናፅ ዘኽእል እምኒ ኩርናዕ ሎም ዕለት ፕረዚዳንት ግዝያዊ ምምሕዳር ትግራይ ኣብ ዝተረከበሉ ከም ዝተነበረ ተሓቢሩ	መጻረይን መወገድ ረሳሕፈሳስን ከተማዋ መቀለን ምሕና ጽእዝ ኽእልእምነኹርና እሎሚያ እለት ፍሬዝዳንቲ ጊዜዋ መምሕዳር ትግራይ ኣብ ሕዝ ተረኽበሉ ከምዚ እተነበረት ሓቢሩ	መጻረይን መወገዲ ረሳሕ ፈሳስን ከተማ መቐለ ን ምህናጽ እዘክእል እምነTከርና ሎም ዕለት ፕሬዚደንት ጊዜያዋ መምሕዳር ትግራይ ኣብ ዝተረኸበሉ ከም ዝተነበረ ተሓቢሩ
Code switching	ሕዚ ናብ meeting ይኸይድ ኣለኹ	ሕዚ ናብ ሚትንቅ ይኸይድ ኣለሆ	ሐዚ ናብ ሚቲንግ ይኸይድ ኣለኩ	ሕዚ ናብ ሚቲንክ ኸይድ ኣሎሖ	Not evaluated
Noisy speech	ሳልሳይ ዓመት ምፍራም ውዕሊ ፕርቶርያ ኣመልኪቱ ሕብረት ኣውሮፓ ሃገራት ውድብ ሕብረትን ኣብ ዘውፀኦ ሓበራዊ መግለፂ ስምምዕ	ኣብሳይ ዓመትሕምፍራን ውዕሊ ሰላም ክሪተርያ ምልቲት ሕብረት ኣውሮጳ ኣግራትይትፍድረትን ማኣብ ዘብጡ ህብራዊ መግለጥ	እይ ዓመትን ፍራን ውዕሊ ሰላም ክሪተር ምልት ብረት ኣውሮፖ ራት ረን ኣብ ዘዋ በራዊ መግለፂ ስ	ጉሰይ ኣነትሕንፍራም ወዲሰላምቲሔርያምንሲእሲ ሕብረዳ ኣውሮታ ኣገራዱቲ ፍግረትን ኣብ ገብቱ ኣሕብራ መግለጥይ ችም	Not evaluated
Raya dialect	ዓሰድዋ ሃመየኺ ቐደማኺ ስጥሕ ደኣነየ ኣቲዮ ኣየዋኪ ብዝሓለፈ ኣግነዮ ነረ እኮ እዩ ሃርፈያ ሕምዋ እሉኒማ ዋእ ሀመይማ ኮንኪ ህዝማ ሕሹኪ ዶ ኣትዮ? ደሓና ኢ	ኣሳድዋ ሓመያኺ ባዳባኺሲ ጠሕዳኣነ ኣቲዮ እይዋኽ ብድኃለፋኣግኒሕዮ ነይረኮ እዩ ሃርፈይ ኣሕምዋብድንማ ሃመይማ ኸን ክኸኻ ሕሽክረት ዮ ተሕናያኸ	ኣሳ ዋ ይያክ ማጥሕ ኣኣትየዋክ ፋግ ነብ ሃርዋ ይ ክረት ዮናያ	ኣረድዋ ሓመያኸ ባዳማ ኸዚ ጠሕላነ ኣቲዮ እየወኸተ ንድሓለፋ ግኒ ዮ ነበረ እኩ እዩ ሓርፈይ ያሕምወ ንማ ወሓሜም ኣኹኽነ ፈኸትኻ ሕሽክረት እዮ ደሕናያኸይ	Not evaluated

Notes on Omnilingual Model:

- Omnilingual ASR was excluded from most evaluation scenarios due to version compatibility issues and dependency conflicts during initial testing phase
- Where Omnilingual results are reported in the main text, this limitation is explicitly noted
- The three-model comparison (MMS, Badrex, Samuael) provides sufficient architectural diversity for comparative analysis